

SUCCESSFUL CLINICAL AUDIO–METADATA ALIGNMENT DOES NOT IMPLY DISEASE TRANSFER: A PAIRING-CONTROLLED AUDIT

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ABSTRACT

Clinical audio models increasingly align recordings with participant metadata or metadata-derived text, but these targets also encode demographics, symptoms, missingness, and cohort structure. We introduce a Pairing-Controlled Transfer Audit that trains matched trajectories under correct pairing, global shuffling, disease-label-preserving shuffling, and a post-hoc label-and-recorded-sex-preserving control, then freezes endpoints for profile correspondence, attribute decodability, and disease prediction after covariate balancing. In UKCOVID, Correct improves profile retrieval and sex decodability over label-only shuffling for AST-6L and OPERA-CT. Preserving label and recorded sex substantially attenuates the sex-probe contrast across AST-6L, OPERA-CT, and HeAR while residual retrieval differences remain. Source-only matched disease gains, sex-preserving comparisons, and target-assisted comparisons remain uncertain. Secondary CODA TB, Cambridge COVID-19 Sounds, and Coswara analyses extend the descriptive pattern without establishing a universal null. Profile-correspondence gain and disease transfer are distinct empirical claims.

Index Terms— clinical audio, audio–metadata alignment, contrastive learning, representation auditing, distribution shift

1. INTRODUCTION

Clinical audio models increasingly use participant metadata not only as prediction-time context, but also as supervision for representation learning. Prior work has used metadata to define contrastive pairs, construct textual supervision, or provide multimodal context for downstream diagnosis [1, 2, 3, 4]. Recent audio-language systems further translate structured metadata into clinical descriptions and align them with frozen or pretrained audio representations [5, 6]. This line of work is appealing because clinical labels are scarce while patient profiles are often available.

The same design creates an identification problem. A contrastive objective specifies that paired modalities should be close, but does not specify which shared attributes should support that correspondence. Clinical metadata is heterogeneous: it can encode disease-related information, symptoms, demographics, missingness patterns, and cohort structure. Thus an apparent alignment gain can reflect profile correspondence, label-conditioned population association, or disease information that transfers after the population changes. COVID audio screening already illustrates how apparent performance can collapse after symptom and demographic adjustment [7], and contrastive objectives are known to admit shortcut solutions when the discrimination task can be solved by non-target features [8]. More generally, explicit multimodal alignment can help or hurt depending on what information is shared across modalities [9].

The remaining identification problem is how the pairing relation should be controlled when clinical metadata retains disease-label and demographic structure. Related work has compared correct and shuffled audio-description trajectories, and clinical vision-language work has used pathology-matched candidates to distinguish instance-level linkage from broader diagnostic similarity [10, 11]. These controls motivate our setting but do not resolve it: a global shuffle removes both the observed participant pairing and the label-conditioned metadata association. We compare Correct (C), Within-label (W), and Global (G) pairing. Within-label assigns metadata from another participant with the same disease label, so $C - W$ asks what the observed participant–profile pairing adds beyond label-conditioned metadata association. $W - G$ measures the effect of retaining that label restriction. A post-hoc control, $W_{y,s}$, additionally restricts donors to the same recorded sex. Disease labels construct these control pairings; they are not included in the metadata text.

Our Pairing-Controlled Transfer Audit follows this matched-trajectory logic. Frozen audio and metadata encoders are shared across arms; the projector initialization, audio order, optimizer schedule, and training duration are matched; only the metadata donor assigned to each audio example changes. The resulting endpoints are frozen and evaluated through three separate questions: did the assigned pairing affect profile retrieval, what attributes became more decodable, and did disease prediction improve after covariate balancing? This separation is important because probe accuracy alone need not imply task use [12]. Here, transfer means evaluation on a covariate-balanced target *within each dataset*, not deployment of one model across datasets. The contribution is an audit of distinct empirical claims, not a claim that metadata alignment is generally ineffective or that the controls isolate causal information components.

2. DATA AND METHODS

2.1. Datasets and evidence status

UKCOVID is the discovery dataset [13]. After objective audio filtering, Standard train contains 20,714 participants. Evaluations use Standard test ($n = 11,119$), matched ($n = 1,814$), and a participant-disjoint matched-long set ($n = 4,196$). Recruitment source alone has COVID AUROC 0.9966 in Standard train and 0.5000 after matching. This motivates testing under the selected change in cohort association; it does not show that a disease classifier causally uses recruitment source. UKCOVID test results informed earlier protocol development, so this is an exploratory audit rather than a locked confirmatory benchmark.

Secondary analyses use CODA TB (1,081 eligible participants) [14], Cambridge COVID-19 Sounds (989) [15], and Coswara (1,701) [16], each with a 200-participant matched target. CODA

uses a fixed 100-pair target from released training data; Cambridge is a strict-COVID reconstruction of its Task-2 cough subset, not an official benchmark replication; Coswara uses model-blind secondary matching but is a post-hoc stress test. These analyses have different evidence status and do not constitute independent confirmations of a null disease effect. Matching balances selected measured covariates, not all sources of bias.

2.2. Frozen encoders and pairing-controlled trajectories

We evaluate three frozen UKCOVID audio backbones spanning different pretraining domains: AST-6L, derived from a general-audio AST; OPERA-CT, pretrained for respiratory audio; and HeAR, pretrained for health acoustics. AST-6L retains six blocks of the 12-layer AudioSet-finetuned AST [17]; the depth was fixed before this audit from independent respiratory-audio comparisons and not from UKCOVID matched performance. OPERA-CT is the released 768-dimensional respiratory contrastive Transformer [18]. Its pretraining includes UKCOVID training audio, making it a domain-pretrained comparator. Participant-disjoint splits in this study do not by themselves verify exclusion from upstream pretraining. AST-6L and OPERA-CT define the original audit, whereas HeAR is reported as a post-hoc extension.

Backbone-specific preprocessing produces one aggregated audio representation per participant. A frozen Phi-2 encoder [19] embeds a deterministic schema of available demographics, smoking status, respiratory history, symptoms, and explicit missing values. Disease labels, participant identifiers, recruitment source, site, country, platform, device, timestamps, and recording artefacts are excluded from this text.

Let a_i, m_i, y_i, s_i denote audio, metadata, disease label, and recorded sex. For arm A , $\pi_A(i)$ is the metadata donor assigned to audio participant i :

$$\begin{aligned} C : \pi_C(i) &= i, \\ W : \pi_W(i) &\neq i, \quad y_{\pi_W(i)} = y_i, \\ G : \pi_G(i) &\neq i, \\ W_{y,s} : \pi_{W_{y,s}}(i) &\neq i, \quad (y_{\pi_{W_{y,s}}(i)}, s_{\pi_{W_{y,s}}(i)}) = (y_i, s_i). \end{aligned} \quad (1)$$

Global imposes no label or sex restriction, although accidental matches can remain. These four assignments define matched training trajectories: the audio examples, optimizer, seeds, and training length are held fixed while π_A changes. We use Δ_{pair} for $C - W$ and Δ_{label} for $W - G$, evaluated separately for each frozen endpoint. These contrasts test pairing constraints; they are not causal estimands or a decomposition into disjoint information components. R denotes the corresponding frozen raw-audio representation.

The $W_{y,s}$ control uses the same 20,714 training participants and rejects self-pairs, singleton strata, and cross-stratum fallback. The five-participant positive/missing-sex stratum supports valid reassignment. Different-person pairs retain identical metadata text in 6.35–6.61% of $W_{y,s}$ pairs and 3.03–3.33% of W pairs. Thus, donor reassignment is not equivalent to changing every semantic target.

2.3. Projector training and frozen endpoint evaluation

Only the audio-side projector is trained. Let $t_i^A = e_{\text{text}}(m_{\pi_A(i)})$ be its assigned frozen text target. For a batch $\mathcal{B}$ of size B , the one-directional objective is

$$\mathcal{L}_A = -\frac{1}{B} \sum_{i \in \mathcal{B}} \log \frac{\exp(\bar{f}_\theta(a_i)^\top \bar{t}_i^A / \tau)}{\sum_{j \in \mathcal{B}} \exp(\bar{f}_\theta(a_i)^\top \bar{t}_j^A / \tau)}, \quad \tau = 0.07, \quad (2)$$

where bars denote ℓ_2 normalization. The projector is trained for five seeds and 500 epochs, using the final checkpoint without epoch selection. Within a seed, arms share initialization, audio batches, and dropout streams; pairing assignments are fixed, and only the metadata donor changes. The final checkpoint is then frozen for all downstream evaluations. This audits a paper-faithful Stage-1 alignment implementation, not the full RespiraMFM downstream Phi-2/LoRA system.

Endpoint 1: correspondence. Source-validation audio queries retrieve metadata profiles. Identical schemas are one semantic target; macro-profile mean reciprocal rank (MRR) prevents frequent profiles from dominating the reported average. Separate post-hoc diagnostics restrict candidates by recorded sex, or by disease label and recorded sex. These restrictions change the retrieval task, not the trained representation; labels define diagnostic candidate sets rather than encoder inputs.

Endpoint 2: decodability. Identically specified regularized probes predict recorded sex, age, symptoms, recording/cohort variables, and disease. Probe performance concerns the specified readout, not causal use by a disease classifier.

Endpoint 3: disease prediction. Metrics are AUROC, negative log-likelihood (NLL), and Brier score. For the primary source-only endpoint, source data select and calibrate readouts before evaluation on matched targets. Matched splits do not select the projector or source-only readout. References include frozen raw audio and metadata only. A fixed multilayer perceptron (MLP) tests one alternative to the linear readout. A separate target-assisted diagnostic trains, selects, and calibrates the readout on matched-long before evaluation on participant-disjoint matched participants. A further diagnostic fits a one-dimensional calibrator on matched-long for transport to matched. Neither diagnostic is a source-only result.

2.4. Uncertainty and analysis scope

Reported 95% bootstrap intervals jointly resample profiles and seeds for macro-profile retrieval, and participants and seeds for probes and disease metrics, retaining arm and seed pairing. Participants, not matched-pair blocks, are the reported disease resampling units. We interpret these intervals descriptively for each comparison, not as simultaneous evidence across all settings. The data-informed analysis sequence is exploratory; pointwise intervals do not resolve uncertainty from analysis selection. An interval including zero is not an equivalence result.

3. RESULTS

3.1. Label-conditioned trajectories differ from global shuffling

Within-label and Global differ even though neither preserves the original donor identity. For UKCOVID AST, source $W - G$ probe AUROC differences are 0.0646 for COVID, 0.0758 for cough, and 0.0647 for recruitment source; the reported intervals exclude zero. On matched participants, the COVID difference is 0.0126 with an interval including zero, while the cough difference is 0.0803. Thus, restricting metadata donors by label changes readout-accessible information even without restoring the original participant pairing. A $C - G$ comparison alone cannot distinguish this label-conditioned association from the additional $C - W$ contrast.

Table 1. UKCOVID paired contrasts with reported 95% intervals. Retrieval uses source validation; sex probes and source-only disease readouts use matched participants. $W_{y,s}$ and UKCOVID HeAR are post-hoc diagnostics. Intervals are interpreted pointwise, not as simultaneous or equivalence statements.

Endpoint and contrast	AST-6L	OPERA-CT	HeAR
Sex AUROC, $C - W$	0.1912 [0.1669, 0.2151]	0.1125 [0.0943, 0.1311]	0.0862 [0.0734, 0.0995]
Sex AUROC, $C - W_{y,s}$	-0.0004 [-0.0153, 0.0147]	0.0067 [-0.0054, 0.0192]	0.0064 [-0.0011, 0.0136]
Sex AUROC, $W_{y,s} - W$	0.1916 [0.1683, 0.2142]	0.1058 [0.0896, 0.1222]	0.0797 [0.0667, 0.0928]
MRR, $C - W$	0.00318 [0.00161, 0.00492]	0.00323 [0.00155, 0.00500]	0.00711 [0.00476, 0.00971]
MRR, $C - W_{y,s}$	0.00164 [0.00001, 0.00335]	0.00099 [-0.00096, 0.00300]	0.00394 [0.00175, 0.00637]
Disease AUROC, $C - W$	0.0062 [-0.0144, 0.0279]	-0.0020 [-0.0191, 0.0168]	0.0029 [-0.0115, 0.0165]
Disease AUROC, $C - W_{y,s}$	-0.0003 [-0.0228, 0.0241]	0.0009 [-0.0169, 0.0196]	0.0130 [-0.0023, 0.0287]
Disease AUROC, $W_{y,s} - W$	0.0065 [-0.0197, 0.0313]	-0.0029 [-0.0219, 0.0164]	-0.0102 [-0.0256, 0.0053]

3.2. Correct trajectories alter correspondence and sex decodability

Table 1 shows positive $C - W$ retrieval estimates across the three UKCOVID backbones; all five seed-level estimates are positive for each primary backbone. This is the manipulation check for the matched trajectories: changing only the assigned metadata donor produced a detectable frozen-endpoint difference. It does not imply complete learning of a clinical profile or exclude other alignment limitations.

For sex decodability, the large $C - W$ contrasts become much smaller when W is replaced by $W_{y,s}$, while $W_{y,s} - W$ remains positive. The sex-probe contrast is therefore highly sensitive to whether the pairing control preserves recorded-sex consistency. Because $W_{y,s}$ also preserves correlated metadata structure and more frequently retains an unchanged semantic profile than W , we do not interpret this comparison as a sex-specific causal decomposition. It also neither establishes equivalence between C and $W_{y,s}$ nor shows that a disease classifier uses sex. Raw audio generally has higher absolute demographic decodability than aligned representations, so a positive $C - W$ contrast should not be read as an improvement over raw audio. Age-probe increases were reported for AST but not OPERA; age was not isolated by an additional pairing control.

Retrieval is different. Residual $C - W_{y,s}$ MRR estimates are positive for all three backbones, with pointwise intervals above zero for AST and HeAR. The AST interval is close to zero at its lower bound (0.00001). Sex-probe attenuation therefore does not establish that sex consistency explains all profile correspondence.

Under same-sex candidate restriction, $C - W$ MRR is 0.00140 [-0.00065, 0.00350], 0.00060 [-0.00143, 0.00256], and 0.00356 [0.00096, 0.00638] for AST, OPERA, and HeAR. Under same-label-and-sex restriction, it is 0.00245 [-0.00001, 0.00499], 0.00094 [-0.00149, 0.00329], and 0.00451 [0.00115, 0.00813]. We interpret contrasts within each candidate rule. Different candidate libraries change the rank distribution, so comparisons across rules do not quantify a fraction of retrieval gain attributable to sex. The HeAR results support residual correspondence under both restricted candidate rules.

3.3. Frozen disease endpoints do not show a consistent transfer gain

Table 2 places the paired contrasts alongside absolute AUROC. $C - W$ disease differences vary in sign across the primary backbones, and Correct is not uniformly above raw audio. The added $W_{y,s}$ comparison also remains uncertain: $C - W_{y,s}$ is -0.0003 for AST, 0.0009 for OPERA, and 0.0130 for HeAR, with all intervals includ-

Table 2. Source-only disease AUROC on UKCOVID matched. R is frozen raw audio. Bracketed values are the reported absolute 95% intervals for $W_{y,s}$.

Backbone	R	C	W	$W_{y,s}$
AST-6L	0.538	0.529	0.523	0.529 [0.504, 0.554]
OPERA-CT	0.577	0.555	0.557	0.554 [0.529, 0.577]
HeAR	0.542	0.545	0.543	0.532 [0.507, 0.557]

ing zero (Table 1). The HeAR interval [-0.0023, 0.0287] still allows a positive effect; it is not evidence of equivalence. Likewise, $W_{y,s} - W$ disease differences are directionally mixed and uncertain. The large change in sex-probe contrasts therefore does not establish a corresponding disease-prediction advantage or disadvantage.

A fixed MLP gives $C - W$ disease AUROC differences of 0.0050 [-0.0154, 0.0270] for AST and -0.0052 [-0.0208, 0.0110] for OPERA. This tests one nonlinear readout; it does not rule out disease information accessible by other readout conditions. Target-assisted $C - W$ estimates are -0.0082 [-0.0337, 0.0177], 0.0287 [-0.0017, 0.0626], and 0.0047 [-0.0278, 0.0408] for AST, OPERA, and HeAR. In particular, OPERA’s interval allows a sizable benefit. No consistent advantage over W or R was established under this diagnostic, but separate intervals do not test an interaction between pairing and readout-training regime.

Across the 12 dataset-backbone settings, all $C - W$ retrieval point estimates are positive, with 11 reported intervals above zero. All 12 sex-probe $C - W$ intervals are above zero. For the primary matched disease contrast $C - W_{y,s}$, five point estimates are positive and seven are negative, with no interval entirely above zero. Across the nine locked secondary settings, these estimates range from -0.0634 to 0.0476. These descriptive counts show a recurring pattern in result shape: correspondence and sex-probe contrasts are more consistently positive than matched disease contrasts in the audited settings. They are not independent replications, pooled effects, or evidence that all disease effects are zero. Diseases, candidate libraries, and analysis histories differ, and the external targets remain small.

3.4. Secondary fusion and probability-scale diagnostics

On UKCOVID matched, adding raw audio to metadata changes AUROC by 0.0028 [-0.0016, 0.0073] for AST and 0.0041 [-0.0006, 0.0087] for OPERA. Metadata plus Correct exceeds metadata plus Within-label by 0.0156 [0.0034, 0.0276] for AST and 0.0021 [-0.0068, 0.0107] for OPERA. Neither comparison establishes

a consistent advantage over metadata plus raw audio. Different interval outcomes do not establish that the two backbone effects differ.

For source-calibrated predictions, the $C - W$ differences in $-NLL$ are -0.0249 $[-0.0443, -0.0069]$ for AST and -0.0176 $[-0.0335, -0.0010]$ for OPERA. Target calibration reduces these gaps to approximately zero with unchanged AUROC. This is consistent with a probability-scale adjustment, not evidence of new ranking information or a unique explanation of the original discrepancy.

4. DISCUSSION AND LIMITATIONS

The central result is not that alignment fails, but that a detectable correspondence gain does not determine its clinical interpretation. $W - G$ demonstrates a difference associated with label-restricted pairing. Adding recorded-sex consistency substantially reduces the sex-probe contrast with Correct, yet this control is not a pure intervention on sex: it also preserves correlated metadata structure and more frequently retains an unchanged semantic profile. Residual profile-retrieval differences remain. These are distinct observations, not a causal decomposition or a participant-identity result. The audit therefore adds information beyond a Correct-versus-Global ablation alone.

Disease prediction must be evaluated separately. The reported disease contrasts do not establish a consistent benefit across backbones, controls, or readout regimes, but several intervals permit beneficial effects. Retrieval gains are therefore insufficient as evidence of disease benefit in this pipeline. Conversely, uncertain disease contrasts do not show that alignment is generally ineffective or that aligned representations contain no usable disease information. Matching also defines a particular target population rather than isolating a purely pathological acoustic signal.

The evidence is exploratory. UKCOVID influenced protocol development; HeAR, $W_{y,s}$, and target-assisted readouts are post-hoc; external targets have limited precision. Pointwise bootstrap intervals do not account for the full analysis-selection process or preserve matched-pair blocks. Upstream pretraining exposure is not resolved by within-study participant separation. The sex-preserving control does not isolate age, other attributes, or their interactions, and may retain additional structure correlated with its conditioning variables. The experiment audits one projector implementation, not a complete multimodal system.

The synthetic study failed its preregistered general-mechanism gates, so we do not claim a universal causal mechanism. A proposed disease-invariant repair was not trained because the cohorts lacked the prespecified support for cross-environment same-label positives and within-environment opposite-label negatives; this is not a negative performance result for that untrained method.

5. CONCLUSION

Conditioning the pairing control on disease label and recorded sex changes the interpretation of a large sex-probe contrast without eliminating all observed retrieval differences. Neither the original nor the expanded pairing comparisons establish a consistent improvement in matched disease prediction, and individual benefits remain compatible with the intervals. Clinical audio-metadata alignment should therefore be assessed through separate correspondence, decodability, and disease-prediction endpoints, with raw-audio references and explicit uncertainty.

Compliance with Ethical Standards

This work is a secondary analysis of de-identified datasets accessed and used under their respective data-use terms. No new participants were recruited and no clinical decisions were made from model outputs.

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